AI FOR HR MANAGEMENT

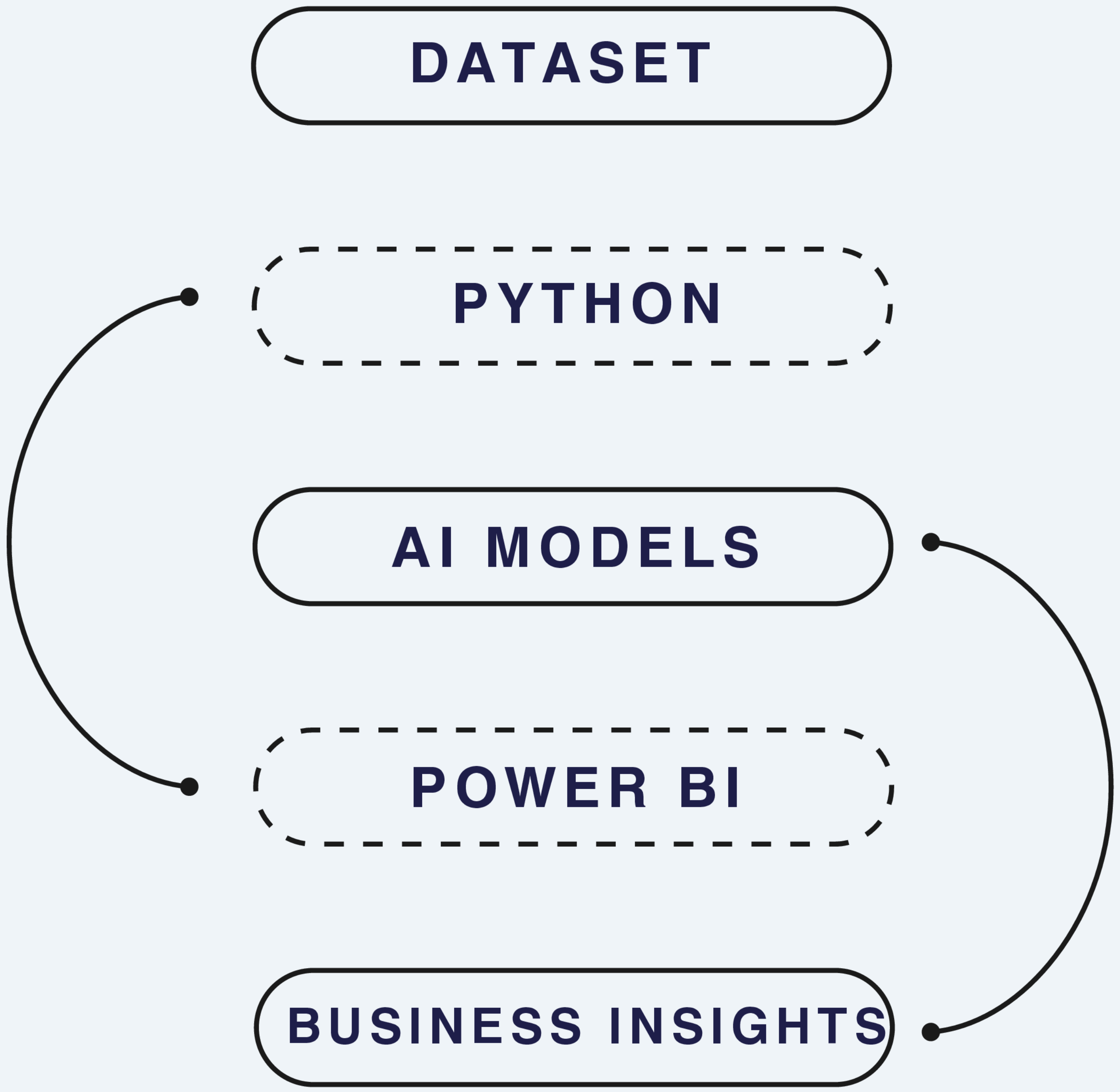
AI-DRIVEN SMART LOGISTICS & SUPPLY CHAIN INTELLIGENCE SYSTEM

Presented by:Zharova Malika,Tokaneva Aisana
AIB-2402

WHAT IS THIS PROJECT ABOUT?

This project analyzes logistics and supply chain data using:

- AI models
- Machine Learning
- Power BI dashboards
- Business Intelligence tools



MAIN GOAL

To improve:

- delivery performance
- customer satisfaction
- operational efficiency
- business decision-making

DATASET DESCRIPTION

Dataset Overview

The dataset contains real-world logistics and supply chain data related to:

- shipments
- orders
- delivery performance
- products
- transportation costs
- financial information

The data was used for:

- AI analysis
- predictive modeling
- customer and shipping analysis
- Power BI visualization

Data Preparation

Before analysis, the dataset was cleaned by:

- removing missing values
- fixing date formats
- converting numerical columns
- preparing data for AI models and dashboards

TECHNOLOGIES USED

Tools & Libraries

- Python
- Google Colab
- Power BI
- Pandas
- NumPy
- Scikit-learn
- Matplotlib

Why These Tools?

- Python was used for AI and machine learning
- Power BI was used for dashboards and visualizations
- Scikit-learn was used to build predictive models

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LinearRegression
from sklearn.linear_model import LogisticRegression
from sklearn.cluster import KMeans

from sklearn.metrics import (
    mean_squared_error,
    accuracy_score,
    classification_report,
    confusion_matrix
)

from sklearn.ensemble import RandomForestClassifier
```

AI MODELS USED

1. Linear Regression

Purpose:

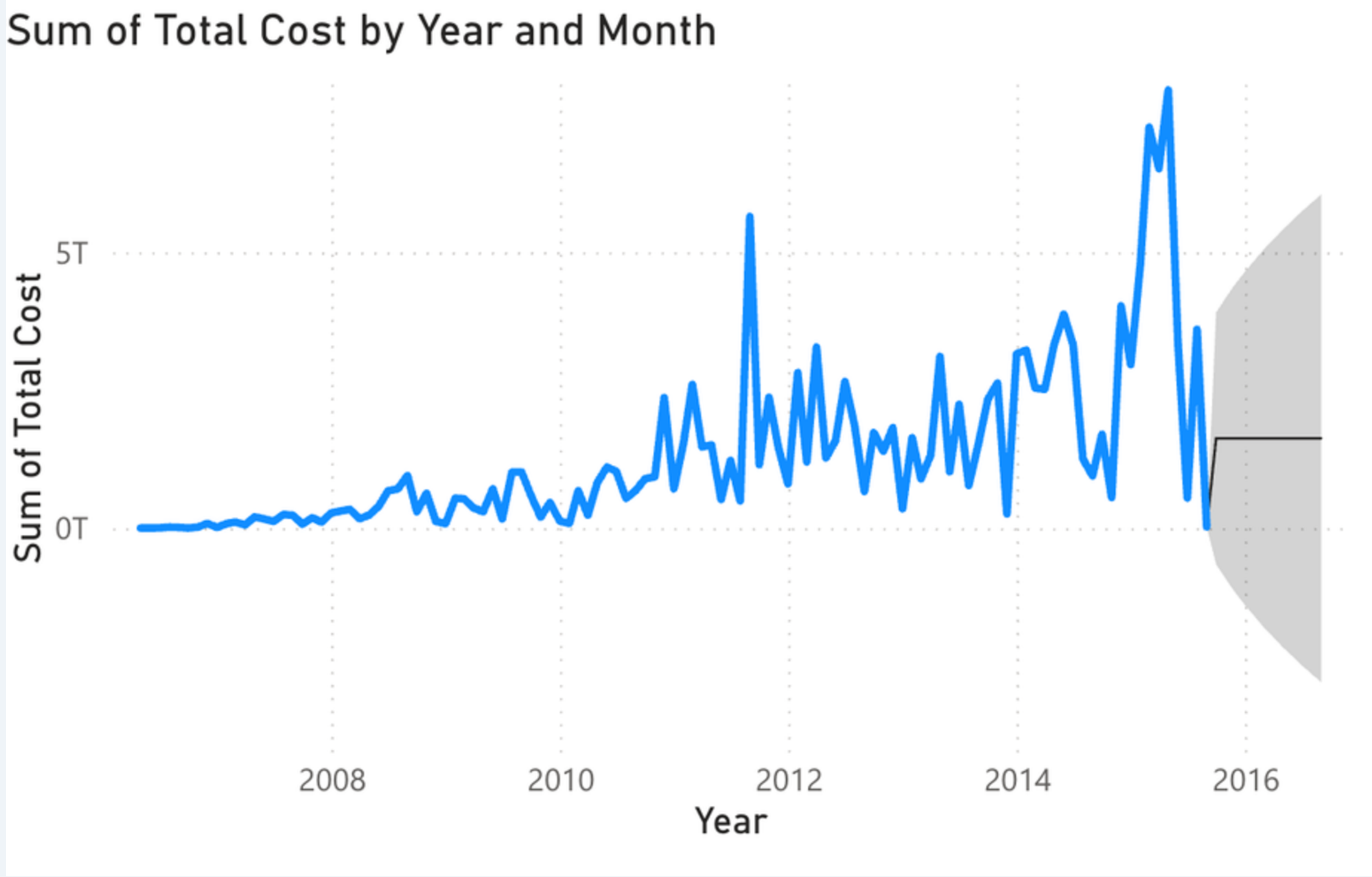
Used in the Demand Forecasting Module to predict Total Cost based on Line Item Quantity.

Why We Chose It:

Linear Regression is simple and effective for predicting numerical values.

What It Found:

Higher order quantities usually increased total shipment cost.



2. Logistic Regression

Purpose:

Used to predict delayed deliveries.

Features Used:

- Shipment Mode
- Country
- Quantity
- Freight Cost

What It Found:

Some shipment conditions increased the probability of delays.

AI MODELS USED

3. Random Forest Classifier

Purpose:

Alternative model for delay prediction.

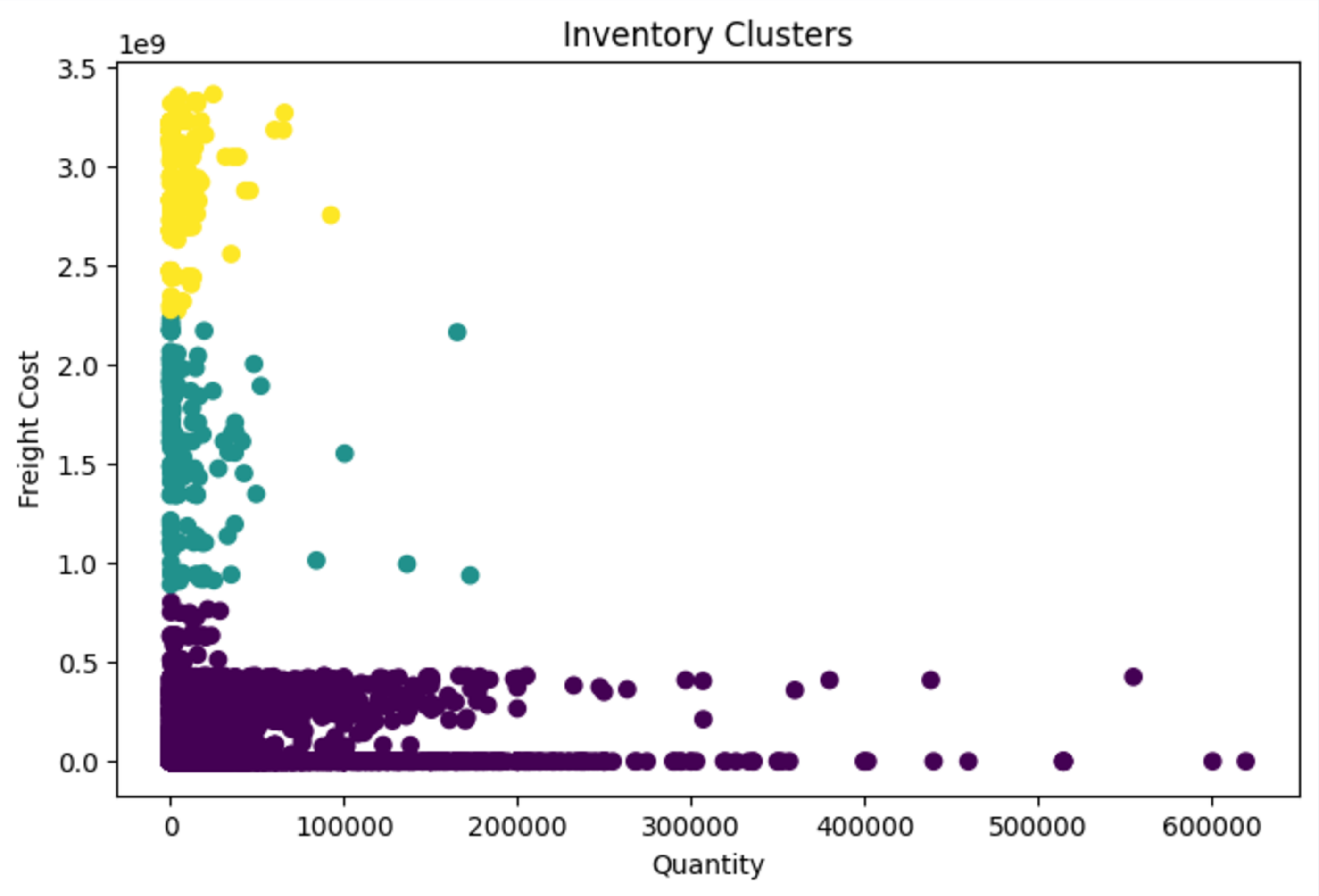
Why We Chose It:

It improves prediction accuracy and handles complex patterns better.

What It Found:

Delivery delays depended on multiple shipment factors.

```
Random Forest Accuracy: 0.9002421307506053
```



4. K-Means Clustering

Purpose:

Used for inventory optimization.

Why We Chose It:

K-Means helps group similar shipment behaviors.

What It Found:

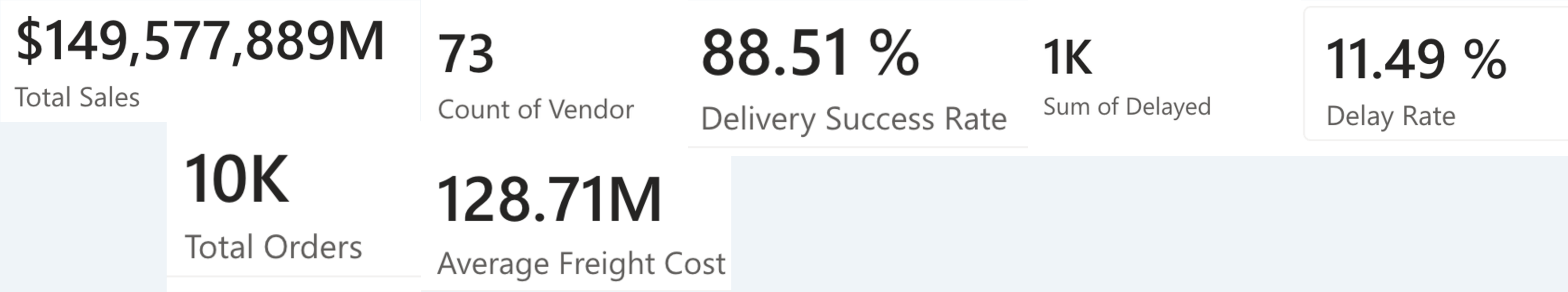
Different inventory behavior groups were identified.

POWER BI DASHBOARD SUMMARY

Main Dashboard Results:

- **Shipment costs were different across shipment modes**
- **Delayed deliveries reduced logistics efficiency**
- **Some suppliers had significantly higher freight costs**
- **Inventory behavior groups were identified using K-Means clustering**
- **Demand forecasting helped predict future shipment costs**

KEY KPIS DISPLAYED



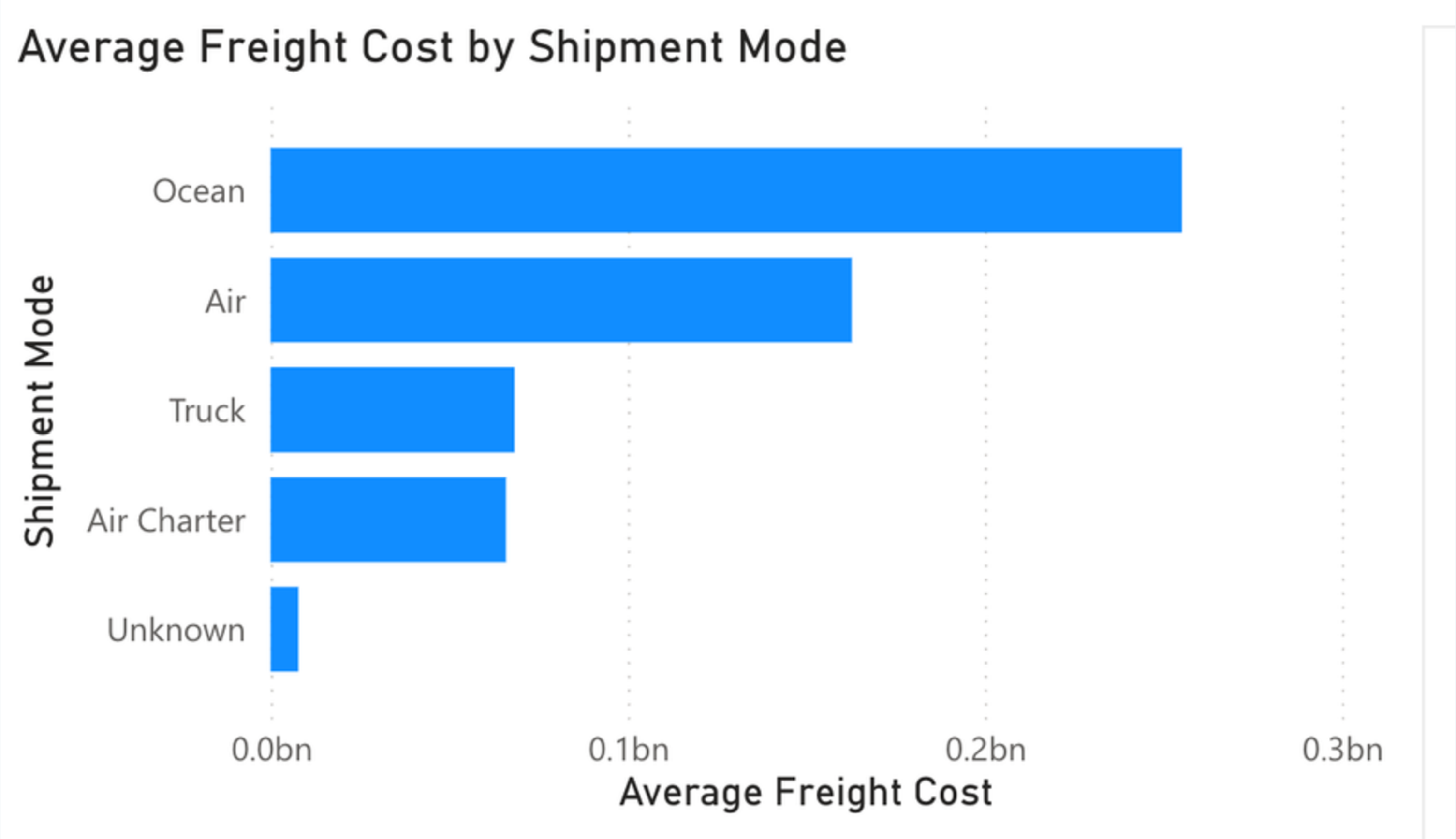
KEY DASHBOARD INSIGHTS

1. Shipment Mode Optimization

The analysis showed that some shipment modes are significantly more expensive than others.
By comparing the average Freight Cost (USD) across different shipment modes, the dashboard helped identify the most costly transportation methods.

Business Impact:

- Companies can reduce logistics expenses
- Managers can choose more cost-effective shipment methods
- Transportation planning becomes more efficient



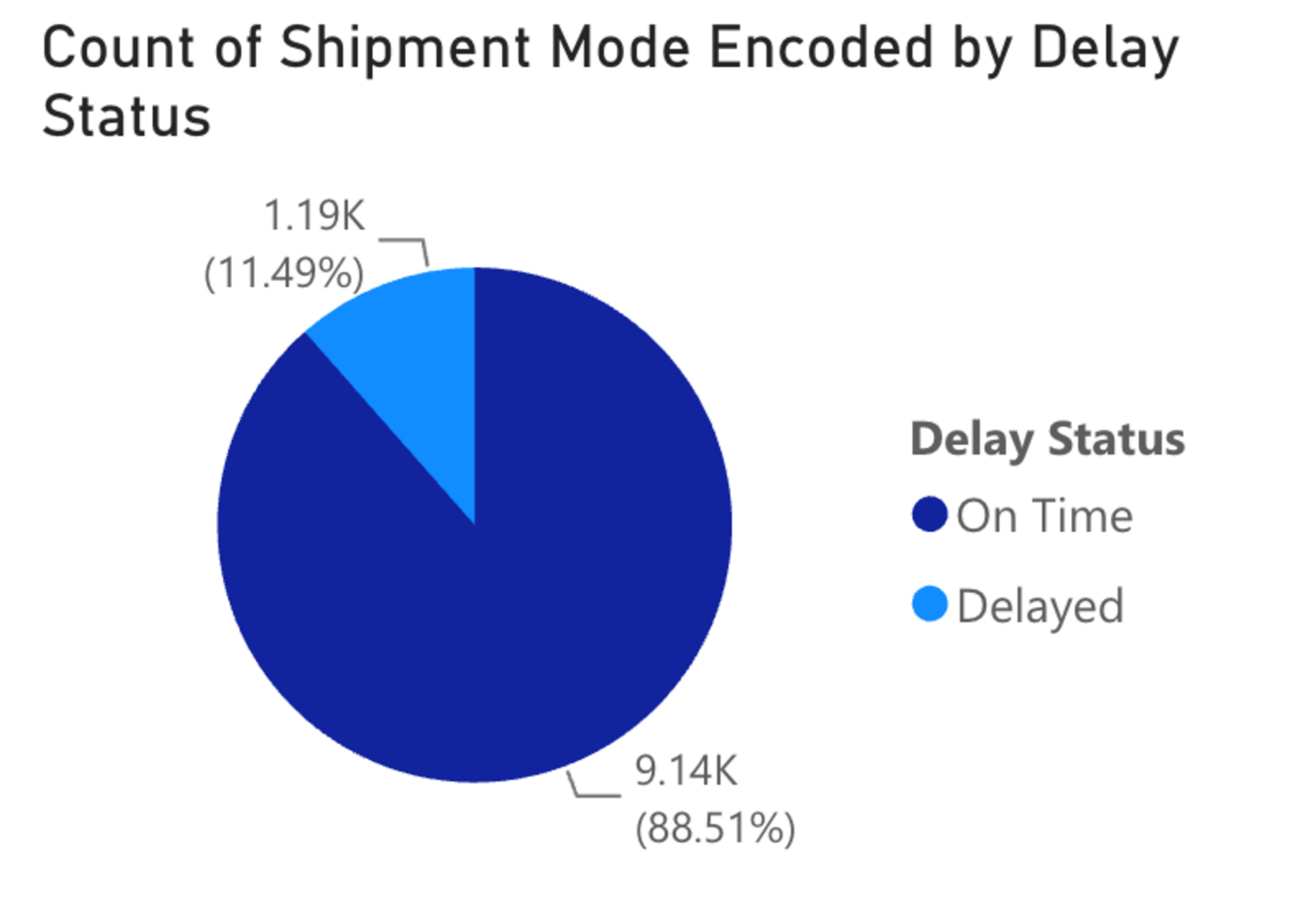
2. Impact of Delivery Delays

The AI prediction models confirmed that delivery delays strongly affect logistics efficiency and operational performance.

Using Logistic Regression and Random Forest models, the system predicted delayed deliveries based on shipment conditions.

Business Impact:

- Better delay prediction
- Improved shipment planning
- Faster operational response
- Reduced supply chain disruptions



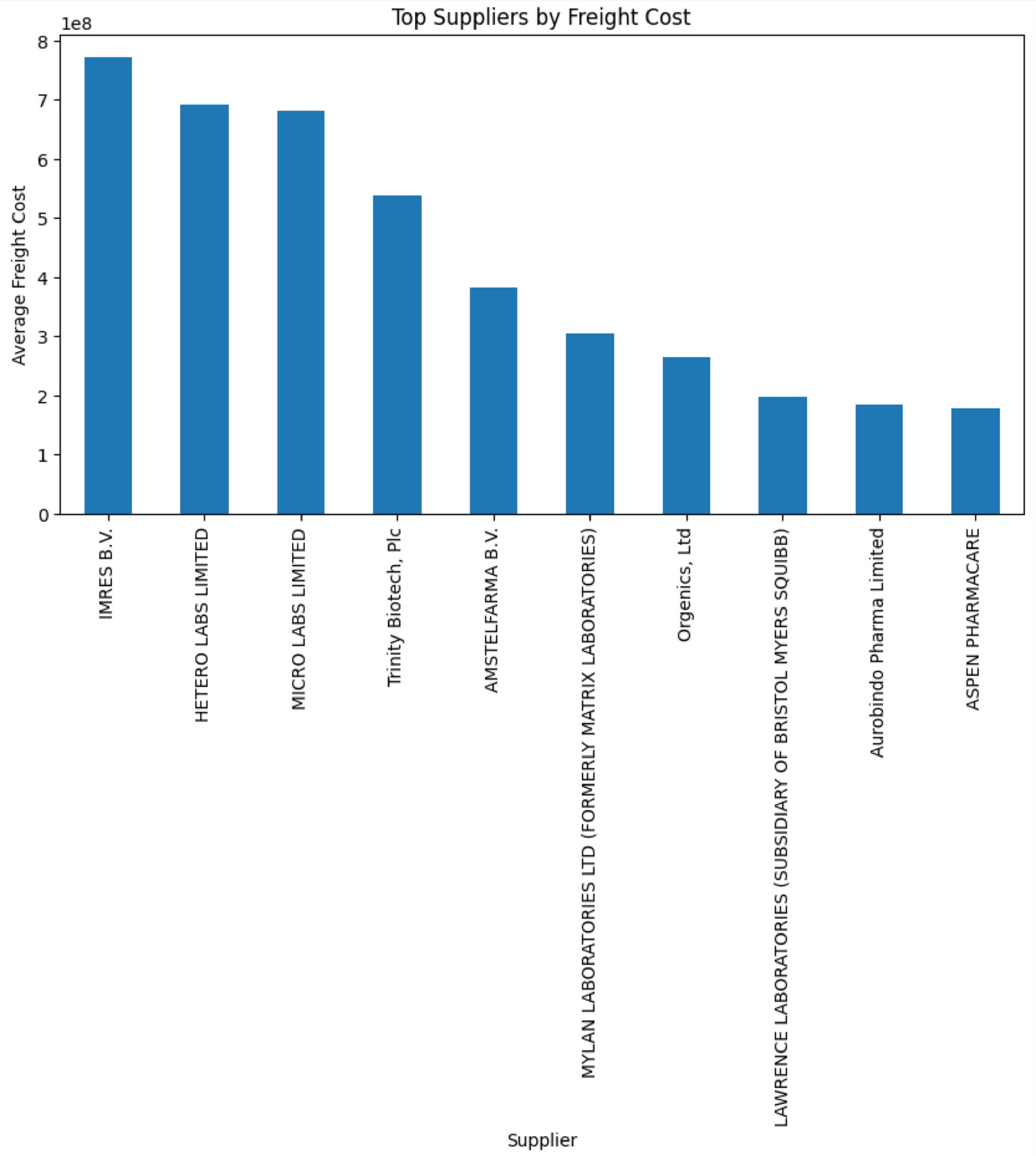
3. Supplier Performance & Cost Management

The supplier analysis revealed that certain vendors generate much higher transportation costs compared to others.

This insight helps companies evaluate supplier efficiency and improve procurement decisions.

Business Impact:

- Better supplier negotiations
- Reduced freight expenses
- Improved cost management
- Smarter supplier selection



THANK YOU!